



## Novel Method for Rapid Screening of Chickpea for Combined Dry Root Rot Disease and Osmotic Stress

Shubhashish Ranjan, Chaitali Narendra Chavan,  
and Muthappa Senthil-Kumar

### Abstract

Chickpea (*Cicer arietinum* L.), confronts substantial challenges from the emerging pathogenic fungus *Macrophomina phaseolina* (Tassi) Goid, causing dry root rot (DRR) disease. Chickpea plants severely affected by combined DRR and drought stress. Currently sick plot and sick pot method are utilized for germplasm screening to identify tolerant genotypes. These methods are time-consuming; therefore, we propose a novel methodology for the rapid screening of chickpea under combined DRR and osmotic stress conditions. This chapter introduces an adept high-throughput phenotyping methodology, conducted within controlled laboratory conditions, aiming to investigate the interaction between osmotic stress and DRR disease in chickpea crops. The methodology employs an innovative pouch technique for screening combined stress, providing a streamlined temporal investigation process and precise control over stress parameters. The incorporation of polyethylene glycol (PEG) enables the simultaneous imposition of osmotic stress alongside pathogen infection, making the methodology versatile for studying combined stress scenarios. This approach fills a gap in concurrent stress imposition techniques, enhancing germplasm screening by identifying genotypes with varying susceptibility and resistance levels. Thus, we suggest use of high-throughput phenotyping in combination genome-wide association study (GWAS) can take combined stress resistance breeding in chickpea at next level to combat food security and climate change.

**Key words** Chickpea, DRR, Combined DRR osmotic stress phenotyping, GWAS and *Macrophomina phaseolina*

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## 1 Introduction

Chickpea (*Cicer arietinum*), an annual legume in global agriculture and nutrition, encounters challenges from both abiotic and biotic stressors that affect productivity [1–3]. The emerging fungal pathogen *M. phaseolina*, causing DRR in chickpea, poses a substantial threat, exhibiting symptoms such as scattered drying of plants in the field, drooping petioles, and compromised lateral and taproots, ultimately leading to severe yield losses, particularly in regions with suboptimal growing conditions [4]. Additionally, this pathogen, a

member of the Botryosphaeriaceae family, demonstrates a broad host range, infecting over 500 plant species, emphasizing its significant implications for chickpea crops [4, 5]. Concurrently, drought or osmotic stress serves as a major constraint, contributing to a global yield reduction of 40–50% in chickpea crops [6, 7]. Osmotic stress during the reproductive stage leads to a considerable decrease in plant yield and related traits [8]. Studies on DRR suggest that osmotic stress influences the interaction between the DRR pathogen and chickpea, exacerbating the severity of DRR disease [9]. The combined osmotic stress and DRR pathogen infection proves to be particularly detrimental to the overall plant performance, resulting in a significant decline in yield. Reductions in yield are more pronounced under the combined stress, surpassing the impact observed under sole pathogen stress [1, 3, 10]. This combined stress condition also leads to significant reductions in photosynthetic rate, transpiration, stomatal conductance, and overall plant growth [9]. Notably, osmotic stress exacerbates DRR, amplifying disease incidence and severity regardless of soil type, geography, and genotype [11, 12]. These findings underscore the importance of understanding and mitigating the impact of combined stress in chickpea cultivation, forming the basis for the high-throughput phenotyping methodology developed in this chapter.

The evolving climatic conditions anticipated to affect the severity of DRR across various crops, thereby raising concerns about cultivating crops in changing environmental condition. This scenario necessitates a more in-depth investigation into the molecular basis of resistance against combined stress and the pathogenicity of *M. phaseolina*. Such insights can be harnessed to formulate more effective control strategies [13]. In addressing the complex dynamics of combined stress, particularly the combat interplay between osmotic stress and DRR in chickpea crops. High-throughput phenotyping studies emerge as crucial factor to understand. Understanding the diverse responses of different genotypes to varying degrees of damage becomes essential for advancements in crop development initiatives. Currently, chickpea screening relies on field-based sick plot and sick pot methodologies, which are time-intensive, laborious, and require substantial infrastructure. Screening a large number of germplasm lines under combined stress conditions within a short time frame is not feasible by using traditional methods. This chapter primarily focuses on establishing a robust methodology for high-throughput phenotyping, specifically directed at assessing the impact of combined stress on chickpea crops. This methodological foundation serves as a crucial stepping-stone toward comprehending the intricate interactions involved in combined stress, offering invaluable insights for future strides in chickpea cultivation and stress resilience. Furthermore, the integration of the high-throughput phenotyping approach with GWAS and developed databases like the Stress Combinations and

Their Interactions in Plants (SCIP) Database [3] offers opportunities for comprehensive research. This integrated strategy streamlines the process of identifying critical genetic and phenotypic markers essential for enhancing crop resilience when confronted with multiple stressors in climate change.

Here phenotyping methodology for combined stress of DRR and osmotic stress simultaneously. This approach, beneficial for high-throughput phenotyping of chickpea germplasm under combined stress, contributes significantly to advancing our understanding of crop responses and resilience under combined stress in chickpea.

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## 2 Materials

- |                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>2.1 Seed Material</b>               | Chickpea seeds [seedbank of International Crop Research Institute for Semi-Arid Tropics (ICRISAT) Patancheru, Hyderabad, Telangana, India].                                                                                                                                                                                                                                                                                                                                                                     |
| <b>2.2 Pathogen</b>                    | Culture of DRR causing pathogen <i>Macrophomina phaseolina</i> (ITCC 8635, NCBI GenBank ID MH509971.1) ( <i>see Note 1</i> ).                                                                                                                                                                                                                                                                                                                                                                                   |
| <b>2.3 Media and Chemical Reagents</b> | <ol style="list-style-type: none"> <li>1. Potato dextrose agar (PDA) (<i>see Note 2</i>).</li> <li>2. Potato dextrose broth (PDB).</li> <li>3. 2% Sodium hypochlorite.</li> <li>4. Polyethylene glycol (mw 20,000).</li> <li>5. Soilrite mix.</li> </ol>                                                                                                                                                                                                                                                        |
| <b>2.4 General Requirements</b>        | <ol style="list-style-type: none"> <li>1. Petri plates for culturing fungus.</li> <li>2. Falcon tubes (50 mL).</li> <li>3. Grow bags.</li> <li>4. 500 mL conical flask</li> <li>5. 500 mL laboratory bottle</li> <li>6. Nylon mesh.</li> <li>7. Petri plates.</li> <li>8. Forceps.</li> <li>9. Autoclaved RO water.</li> <li>10. Polythene ziplock bags (5 × 6 in.).</li> <li>11. Plastic tray.</li> <li>12. Wooden sticks.</li> <li>13. Paint brush.</li> <li>14. Scalpel.</li> <li>15. Test tubes.</li> </ol> |

16. Test tube stands.
17. Milli-Q (MQ) and reverse osmosis (RO) water.

## 2.5 Instruments

1. Laminar airflow.
2. Incubator (28 °C).
3. Shaker incubator (28 °C).
4. EPSON root scanner.
5. Weighing balance.
6. Electrolyte conductivity (EC) meter.

## 2.6 Software

RhizoVision [14], WinRHIZO (Regent Instruments Inc., Canada), ImageJ [15], or other similar image analysis tools (*see Note 3*).

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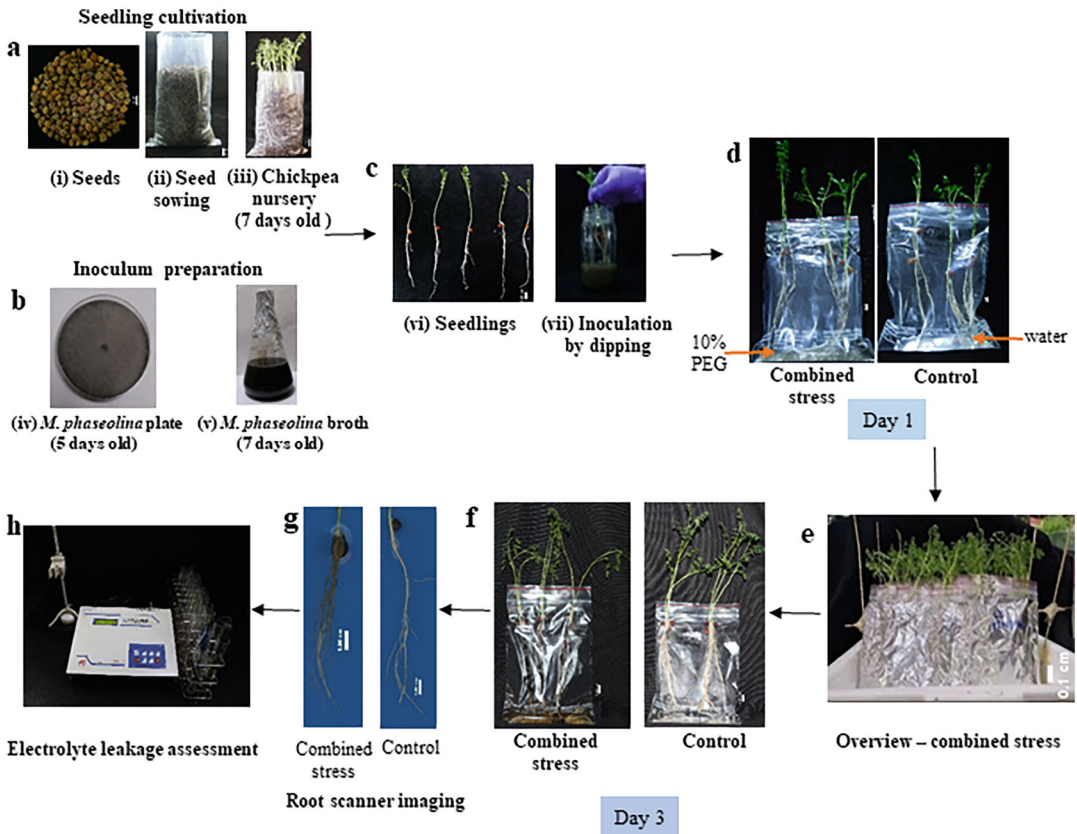
# 3 Methods

## 3.1 Seedling Cultivation

1. Chickpea seeds (40) per genotype wash with autoclaved RO water under laminar airflow, repeating the process 3 to 4 times.
2. Surface sterilize the seeds using 2% sodium hypochlorite, shaken intermittently for 3 min, and rinsed with autoclaved RO water 3–4 times to remove traces of sodium hypochlorite.
3. Fill autoclaved soilrite into grow bags, sow surface-sterilized seeds in the bags and maintain at 25 °C with a 16-h light and 8-h dark photoperiod, and a light intensity of 150  $\mu\text{mol}/\text{m}^2/\text{s}$  in growth room or chamber (Fig. 1).
4. Uproot the plants 7 days after sowing. Gently wash the plants to remove soil particles without damaging the roots, and place in a tray with autoclaved RO water.

## 3.2 Fungal Inoculum and 10% PEG Preparation

1. Prepare 200 mL of PDB media in a 500 mL conical flask and autoclave.
2. Inoculate the medium with a 1 mm agar plug from a 5-to-6-day-old actively growing *M. phaseolina* fungal agar plate.
3. Incubate the conical flasks at 28 °C in a shaker for 7 days at 180 rpm in darkness.
4. Filter fungal mycelia and microsclerotia using nylon mesh after 7 days (*see Note 4*).
5. Weigh 10 g of PEG with a molecular weight of 20,000 (*see Note 5*).



**Fig. 1** Experimental workflow for imposition of combined DRR and osmotic stress in chickpea. (a) Seedling cultivation—(i) chickpea seeds, (ii) seeds sown in a bag containing soilrite mix for germination, and (iii) 7-day-old seedlings. (b) Fungal inoculum preparation—(iv) 5 day-old *M. phaseolina* plate for broth inoculation and (v) broth culture of *M. phaseolina* for root inoculation. (c) Dipping method—(vi) washed chickpea seedlings used for fungal inoculation. (vii) Inoculation of fungus on chickpea roots by dipping method. (d) Fungus inoculated and control seedlings placed in pouch containing 10% PEG and water, respectively. (e) Experimental setup in a tray, maintained in the infection room for 3 days. (f) Combined stress and control seedlings after 3 days after inoculation. (g) Root scanner images of combined stress and control seedling after 3 days of infection. (h) Electrolyte leakage assessment by using an EC meter

6. Dissolve 10 g of PEG 20,000 mw in 60 mL of MQ water in laboratory bottle. Adjust the final volume to 100 mL with MQ water and autoclave.
7. Store the 10% PEG solution at 4 °C or room temperature.

### 3.3 Pouch Preparation

1. Fill 20 mL of autoclaved RO water in bags for control treatment.
2. Fill 20 mL of 10% PEG solution in bags for stress imposition (Fig. 1) (see Notes 6 and 7).

### 3.4 Fungus Inoculation by Dipping

1. Filter the 7-day-old fungal culture using a nylon mesh with a 1 mm diameter.
2. Dissolve the washed mycelia in autoclaved RO water within an autoclaved petri dish.
3. Dip the plant roots into the inoculum for 1 min, moving intermittently to ensure even mycelia attachment (Fig. 1) (*see Note 8*).

### 3.5 Combined Stress Imposition

1. Place the control, osmotic stress, and combined stress (fungus-inoculated roots in 10% PEG medium) in separate bags, using different bags for each genotype.
2. Place plants in prepared respective bags with partial root immersion in water and 10% PEG solution.
3. Cover all bags with aluminum foil to ensure darkness. Leave the cut part of the bag open for air circulation to prevent excess moisture inside.
4. Arrange the bags in a tray with wooden sticks for support. Maintain the setup at 25 °C for 3 days in the infection room under 150  $\mu\text{mol}/\text{m}^2/\text{s}$  light intensity, 60–70% relative humidity (Fig. 1) (*see Note 9*).

### 3.6 Assay for Quantification in Combined Stress

#### 3.6.1 Root Cleaning

1. Remove all plants from the bags after 3 days of incubation.
2. Wash the roots thoroughly in water to eliminate fungal mycelium and PEG from combined and osmotic stress samples.

#### 3.6.2 Root Scanner Imaging and Scoring

1. Visually observe the roots for necrosis, tap and lateral root damage or loss.
2. Impute the observation in provided range of manual score (Table 1) for each plant of genotype.
3. Calculate average score of genotype for manual score.
4. After manual scoring, capture root image using root scanner.
5. Assess these root images using Image J or WinRhizo software, assigning scores on a scale from 0 to 9 to reflect the severity of necrosis in the root region. Additionally, conduct manual scoring, also on a scale from 0 to 9 (as shown in Table 1), for the analysis of infected or diseased roots.
6. To calculate the total area of necrosis from root scanner image, use the following formula:

$$\text{Area of necrosis (\%)} = \left( \frac{\text{Area covered by specific score}}{\text{Total root area}} \right) \times 100$$

7. Calculate average necrosis area from all samples of a genotype to obtain total area of necrosis.

**Table 1**  
**Disease severity for combined DRR under osmotic stress of chickpea**

| Rating    | Symptoms                                                                                                                    | Disease reaction     |
|-----------|-----------------------------------------------------------------------------------------------------------------------------|----------------------|
| 1         | No infection on roots                                                                                                       | Immune               |
| >1 and <3 | Few small lesions on infected roots, only little effect of drought and disease appears on roots                             | Resistant            |
| >3 and <5 | Small and clear lesions on roots; new and lateral roots infected                                                            | Moderately resistant |
| >5 and <7 | Lesions on roots; many new roots generally have lesions, discoloration of roots, lateral and tap roots, root started drying | Susceptible          |
| >7        | Roots are fully infected and completely discolored; sometimes complete roots dried                                          | Highly susceptible   |

### 3.6.3 Electrolyte Leakage Assessment

1. After imaging, roots were cut into small pieces using a scalpel, treating each root as a replicate.
2. Collected roots for each genotype under control, osmotic, and combined stress conditions.
3. Weigh 100 mg of roots from each replicate into labeled test tubes and arrange the test tubes properly in a test tube stand.
4. Add 15 mL of MQ water to each test tube and wash for 30 min.
5. After 30 min, remove the water and add 15 mL of fresh water to each test tube to measure electrolyte leakage (*see Note 10*).
6. Record the EC using an EC meter immediately within 30 min after adding water (ECa), after 4 h of rehydration (ECb), and ultimately after autoclaving to measure total leakage (ECt).
7. Express the measured electrolyte leakage as a percentage electrolyte leakage using the formula.

$$EC (\%) = \left[ \frac{(ECb - ECa)}{ECt} \right] \times 100$$

### 3.6.4 Genotype Screening

1. Sow seeds of chickpea genotypes in a soilrite-containing bag.
2. Uproot the 7-day-old sown plants.
3. Wash the plants and keep in tray by labeling genotype name.
4. Prepare the experimental setup (refer to Subheadings 3.2, 3.3, and 3.4) for combined and osmotic stress imposition.
5. Remove the 3 days postinoculated combined and osmotic stress plant and wash.

6. Measure the manual score (Table 1) and record the image for area of necrosis scoring.
7. Calculate area of necrosis to obtain total area of necrosis.
8. Measure EC from the root samples and compute in the EC formula (*see* **Notes 11** and **12**).
9. Obtain all readings and analyze for combined and osmotic stress.

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## 4 Notes

1. The necrotrophic fungus present in the soil is derived from the roots of a diseased plant harvested from a field in NIPGR, New Delhi (28.5307 °N, 77.1665 °E). A pure culture is successfully cultivated using the hyphal tip technique (Irulappan and Senthil-Kumar 2021).
2. It is preferable to use PDA and PDB from Becton, Dickinson and Company, Sparks, MD, USA for the experiment.
3. RhizoVision, WinRHIZO, Image J, or other similar tools used to analyze necrotic lesions from infected root images for high-throughput phenotyping. These tools help quantify and evaluate the extent of root damage and providing detailed information on the impact of infections.
4. Ensure thorough washing of mycelia with autoclaved RO water to remove all media and toxins.
5. At PEG concentrations above 15%, the roots may become either dry or soggy, depending on the genotype. Therefore, 10% PEG concentration recommended for optimal root phenotype.
6. PEG facilitates the imposition of osmotic stress by offering adjustable concentrations, thus allowing for customization according to specific experimental requirements. This adaptability in concentration selection ensures precise control over the intensity of osmotic stress applied, thereby enhancing the flexibility and effectiveness of experimental procedures.
7. Cut bags slightly on both sides of upper part to facilitate proper air circulation and preventing root soggianness due to excess moisture.
8. After loading fungal inoculum, place seedling roots on tissue paper to absorb excess water from the inoculum, thereby preventing soggianness in the roots.
9. Optimize environmental conditions are necessary for optimal disease expression and to obtain phenotype. High temperature (above 30 °C) and high light intensity may cause drying of

roots, whereas lower temperature (below 22 °C) and high relative humidity may cause sogginess in roots. At 20 °C and below temperature, no fungal infection may be observe.

10. Maintain a washing time of 30 min to ensure accurate observation of the EC trend within the same genotype root samples. A shorter washing time may result in unclear EC readings. Additionally, calibrate the EC meter regularly to ensure precise measurements for proper experimentation.
11. Using this protocol, 25 genotypes/week/m<sup>2</sup> can be screen. The responses of genotypes categorized as resistant or susceptible across all three methods can be used to classify them as either resistant or susceptible to combined DRR osmotic stress.
12. Chickpea genotypes ICC4958 and ICC11088 (BG212) recommended as check genotypes for combined stress experiments. ICC4958 is moderately resistant, with a manual score of 4.5, an area of necrosis of 37%, and an EC of 25.5%. On the other hand, ICC11088 is highly susceptible, with a manual score of 8.4, an area of necrosis of 55.02%, and an EC of 45%. These genotypes can serve as benchmarks for evaluating stress responses using this protocol.

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